

QuantROI System Architecture

High-Level Architecture Overview



Blockchain Layer

Solana Smart Contracts

Delegation Management Payment System RIA Compliance

Founder Auth Auto-payment Audit Trail

Duty Record Milestone Pay SE

Oracle Batch Contract Terms Co

Knowledge ZKP Strategy Verification AI Auditor Proof Verify Completeness Strategy Val Sincerity Risk Mgmt

Data Layer

Neo4j Redis IPFS

Graph Database Cache Distributed Storage

Causal Nodes Oracle Cache Immutability

Vote Nodes Query Cache Hash St

News Nodes Session Data Audit T

Relationships <100ms Access ZKP Pro

Infrastructure Layer

Kubernetes Cluster

ArgoCD Prometheus Datadog

GitOps Monitoring Observ.

MLflow Kafka Load Bal.

Hierarchy Streaming Nginx

Component Architecture Details

1. Oracle Optimization Architecture

Oracle Optimization System

Oracle Provider Layer

MUTUALLY EXCLUSIVE OPTIONS (Either/Or, Not Both)

Option A: Supra Oracles Option B: Audited Libraries

â€¢ Zero-block delay â€¢ SEC-compliance

â€¢ <1s data feeds â€¢ Verified inputs

â€¢ High-frequency updates â€¢ Audit trails

â€¢ Consensus pricing â€¢ Enterprise-grade

Enhancement Components

Chainlink VRF Redis Cache Polygon Miden

â€¢ Verifiable â€¢ <100ms resp â€¢ ZK-Rollups

â€¢ Randomness â€¢ TTL mgmt â€¢ Local fallback

â€¢ ZKP voting â€¢ Hit rate opt â€¢ Proof verification

Solana Batch Processing

â€¢ BatchOracleRequest struct for multiple verifications

â€¢ Single transaction for multiple oracle calls

â€¢ 80% success rate requirement with latency monitoring

â€¢ Event emission for audit trails and performance tracking

Performance Monitoring

â€¢ Sub-second finality tracking (target: <1000ms)

â€¢ Cache hit rate optimization (target: >80%)

â€¢ Oracle uptime monitoring (target: 99.99%)

â€¢ Latency distribution analysis

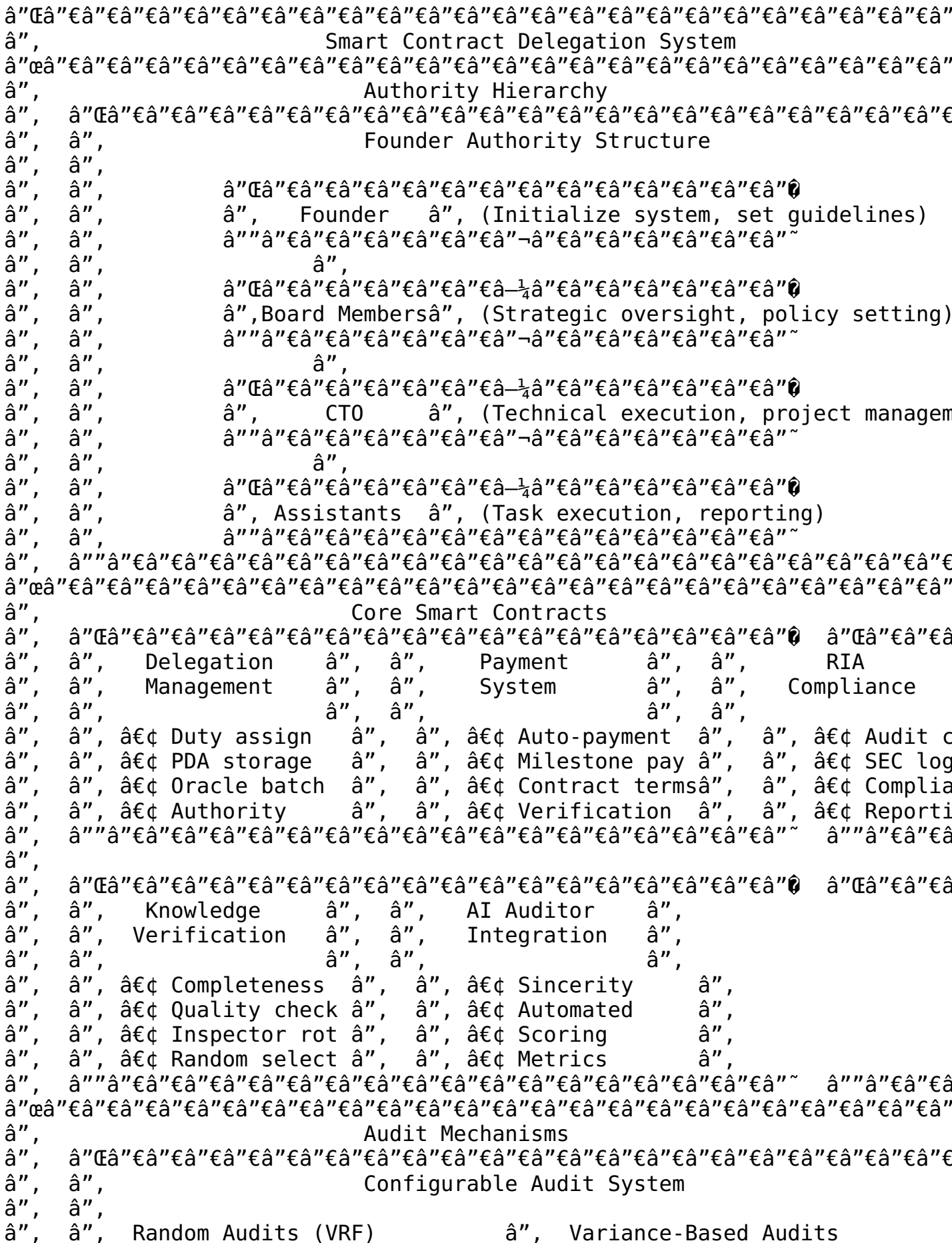
â€¢ Automatic failover and consensus pricing

2. ZKP Voting System Architecture

ZKP Voting Pipeline

Input Layer									
User Vote		RL Vote ID		Oracle Data					
Suggestion		Dynamic Gen		Market					
Confidence		Causal Links		Verific					
Symbols		ML Enhanced		Sub-sec					
Processing Layer									
ZKP Proof Generation									
Circom		Groth16		Poseidon					
Circuit		Protocol		Hash					
Vote anon		Proof gen		Cr					
Input valid		<2s target		Co					
Oracle-Enhanced Verification									
Market data validation through optimized oracles									
Sub-second verification with cached responses									
Miden ZK-rollup integration for faster proof verification									
Chainlink VRF for verifiable randomness in vote selection									
Storage Layer									
Neo4j		IPFS		Solana					
Vote Nodes		Proof Storage		Smart Contract					
Vote metadata		Immutable		On-chain					
Relationships		Hash verify		Event e					
Query index		Distributed		Audit t					
Security & Compliance									
Delayed vote detection for manipulation prevention									
ZKP proof verification with cryptographic guarantees									
Oracle data integrity through consensus pricing									
SEC compliance logging with immutable audit trails									
Performance monitoring for sub-second finality requirements									

3. Smart Contract Delegation Architecture



Throughput Requirements

- **Solana TPS:** 65,000 transactions per second
- **Event Ingestion:** 20,000 events per second
- **Concurrent Users:** 100,000 users
- **Oracle Calls:** 1,000+ per second with batching
- **Database Queries:** 10,000+ per second with caching

Availability Requirements

- **System Uptime:** 99.999% (5.26 minutes downtime/year)
- **Oracle Uptime:** 99.99% with automatic failover
- **Database Availability:** 99.95% with replication
- **API Availability:** 99.99% with load balancing

Security Architecture

Cryptographic Security

- **Quantum-Resistant:** CRYSTALS-Kyber/Dilithium
- **Hash Functions:** SHA-3 for audit trails
- **ZKP Protocols:** Groth16 with Poseidon hash
- **Digital Signatures:** Ed25519 for Solana transactions
- **Encryption:** AES-256-GCM for data at rest

Access Control

- **Multi-Factor Authentication:** Biometric + quantum-resistant
- **Role-Based Access Control:** Hierarchical permissions
- **Smart Contract Permissions:** Authority-based execution
- **API Authentication:** ZKP-based client verification
- **Audit Logging:** Immutable access trail recording

Compliance Architecture

Regulatory Compliance

- **RIA Internet Exception:** Automated compliance with human oversight
- **SEC Rule 10b-5:** Comprehensive disclosure requirements
- **SEC Rule 204-2:** Recordkeeping with 4 hours/year requirement
- **SOC2 Type II:** Automated controls and monitoring
- **GDPR:** Data privacy and right to erasure

Audit Trail Architecture

- **Immutable Logging:** SHA-3 cryptographic hashing
- **Distributed Storage:** IPFS for tamper-proof records
- **Real-time Monitoring:** Prometheus + Datadog integration
- **Compliance Reporting:** Automated SEC-compliant reports

- **Bias Monitoring:** <0.1 threshold for ethical AI compliance
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Architecture Review: Technical Implementation Guide

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